

Cross-lingual offensive speech identification with transfer learning for low-resource languages[☆]

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ABSTRACT

Most of research on the identification of offensive speech on social media platforms exist in English and other rich languages. A series of recently proposed methods for detecting low-resource offensive languages require labeled data. In this work, we propose an unsupervised model that can detect offensive speech for low-resource languages. Our method does not depend on any labeled data of low-resource languages. In detail, we propose an agreement regularized training that combines adversarial learning and transfer learning. Augmenting low-resource training data with sample regeneration methods to maintain the performance of the trained offensive speech identification model from rich-resource to low-resource languages. Extensive experiments on four low-resource languages demonstrate that our model either is on par or outperforms the supervised methods, without employing any annotated data on real-world offensive speech detection tasks for low-resource languages.

1. Introduction

The presence of hate speech, cyber-bullying, cyber-aggression, and other forms of offensive language in online communities is a global phenomenon, on such a scale that its automatic detection is the subject of many studies [1–3]. Recent many neural network methods have achieved obvious performance on offensive speech identification in English and a few of other main languages (such as French et al.) [4–8]. Although those methods can succeed, they heavily depend on the availability of a large amount of labeled data. Low-resource languages¹ often have limited labeled data or even without any labels, which may make us disable to train a robust and accurate offensive speech identification model. To build an offensive speech identification model for low-resource languages, recently many researchers used cross-lingual word embeddings to solve the data scarcity problem for low-resource languages [9–13]. The core of those methods is to transfer knowledge from a resource-rich (source) language² to a low-resource (target) language. Ranasinghe and Zampieri [13] first used contextual embeddings to solve the offensive speech identification for low-resource languages. They, however, still required a certain amount of labeled data of low-resource languages. Such a limitation may prevent these models from being applicable in some low-resource languages with a very small amount of labeled data even without any labeled data.

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¹ Low-resource languages include languages or domain-specific with sparse texts.

² Languages and some domains with large amounts of related texts are called resource-rich languages.

Recently, a strong point is that continuous word representation spaces exhibit similar structures across languages, even when considering distant language pairs like English and Chinese. If we can exploit the similar assumption by learning a mapping function³ that projects a source and target embeddings into a shared space in an unsupervised way, then we can directly transfer the trained offensive speech identification model on a resource-rich (source) language into a low-resource (target) language, and call this process cross-language mapping. However, there are two problems that should be addressed: (i) how to learn a mapping function that project two language embeddings into a shared space with an unsupervised way. (ii) how to hold the performance of the trained offensive speech identification model when implementing transfer learning⁴ from rich-resource to low-resource languages. To address the first issue, we present two effective approaches to map two language embeddings into a shared space: adversarial training and Procrustes analysis. Adversarial training⁵ constructs pseudo-aligned shared space for two language embeddings. Although we can use adversarial learning to obtain an approximate alignment, many functional words may mistake the alignment [14]. We further proposed Procrustes analysis that uses the term frequency-inverse document frequency (TF-IDF) to obtain a fine-tune alignment.

For the other problem, most previous methods [8,10,11] only independently map two language embeddings into a shared space and implement transfer learning, which ignores the shared representations for specific tasks when mapping a shared space. To address this problem, we propose an effective agreement regularized training to regularize and update the parameters of trained offensive speech identification model for transfer learning in detecting low-resource language offensive speech. The training process is that we first use the adversarial training and Procrustes analysis to map two monolingual word embeddings (rich-resource and low-resource languages) into a shared space. Then, we use the shared representations to train offensive language identification model in rich-resource language. For offensive speech identification of low-resource languages, we first calculate a most similar sentence of low-resource language with labeled data of rich-resource language. And we treat the selected sentence as labeled data of low-resource language. We again train the trained offensive speech identification model use new labeled data of low-resource language and previous labeled data of rich-resource language. This regularized training can hold the performance of a trained offensive speech identification model in low-resource languages.

Our main contributions of this paper are the following:

- (1) We present an offensive language identification model without any labeled data of low-resource languages and build our model that reaches or outperforms previous supervised methods on shared evaluation tasks.
- (2) In terms of model training, we introduce an agreement regularized training schemas that update the trained offensive speech identification model considering low-resource and rich-resource languages at the same time. This can encourage the model to make better use of rich-resource languages.
- (3) We demonstrate the effectiveness of our method on four low-resource languages. We also make the code⁶ and the models freely available to the community.

The rest of this paper is structured as follows: Section 2 reviews the methods of identifying online offensive speech and points out the shortcomings of these methods. Section 3 introduces the problems faced by using the transfer learning method to identify cross-lingual offensive speech for low-resource and the methods proposed to solve these problems. Section 4 introduces the equipment and data required for the experiment. Then, experiments are carried out in Section 5 to prove the effectiveness of this method. Finally, Section 6 summarizes the method of identifying low-resource cross-lingual offensive speech proposed in this paper.

2. Related work

It is a growing interest in the development of computational models to identify offensive content online [1,2,15]. Recent many approaches that relied on deep learning heavily need amount of labeled data, a substantial portion of these studies consider English or only a few main languages [4]. However, many languages face the problem of offensive data scarcity [16,17]. To address this problem, many researchers used recently prevalent cross-lingual word embeddings to implement offensive speech identification in low-resource languages [18]. The basic process is training a classification model of offensive speech identification on rich-resource language. Then, using transfer learning to transfer the trained model from a rich-resource language to a low-resource language. The core of those methods is how to learn a shared embedding space that is indicative of classification for both languages. The language-invariant is the foundation of transfer learning [19]. However, they still require a certain amount of labeled data. This may hinder these models from being applicable in some low-resource languages with a very small amount of labeled data even without any labeled data.

Recently, there have been several works at developing “zero-resource” methods to solve natural language processing classification [14,20,21]. Moreover, [19] proposed a cross-lingual unsupervised method to implement sentiment classification task. This method use transfer learning that builds on pivot-based unsupervised machine translation. Their method achieves a significant performance in cross-lingual sentiment classification. We believe that the “zero-resource” methods also can be applied to offensive speech identification for low-resource languages. Our proposed method is inspired by them. This paper, propose an unsupervised method that can detect offensive speech for low-resource languages. To the best of our knowledge, however, unsupervised cross-lingual methods have not yet been applied to offensive speech identification.

³ The basic idea of cross-language mapping is to train embeddings in monolingual corpora using different languages independently, and then map them to a shared space through a linear transformation.

⁴ The parameters of the trained model (pre-trained model) are transferred to the new model to help the new model training.

⁵ Adversarial training is a training method that introduces noise, which can regularize parameters and improve model robustness and generalization ability.

⁶ <https://github.com/zsl-nlp/DeepOffense-Unsupervised.git>.

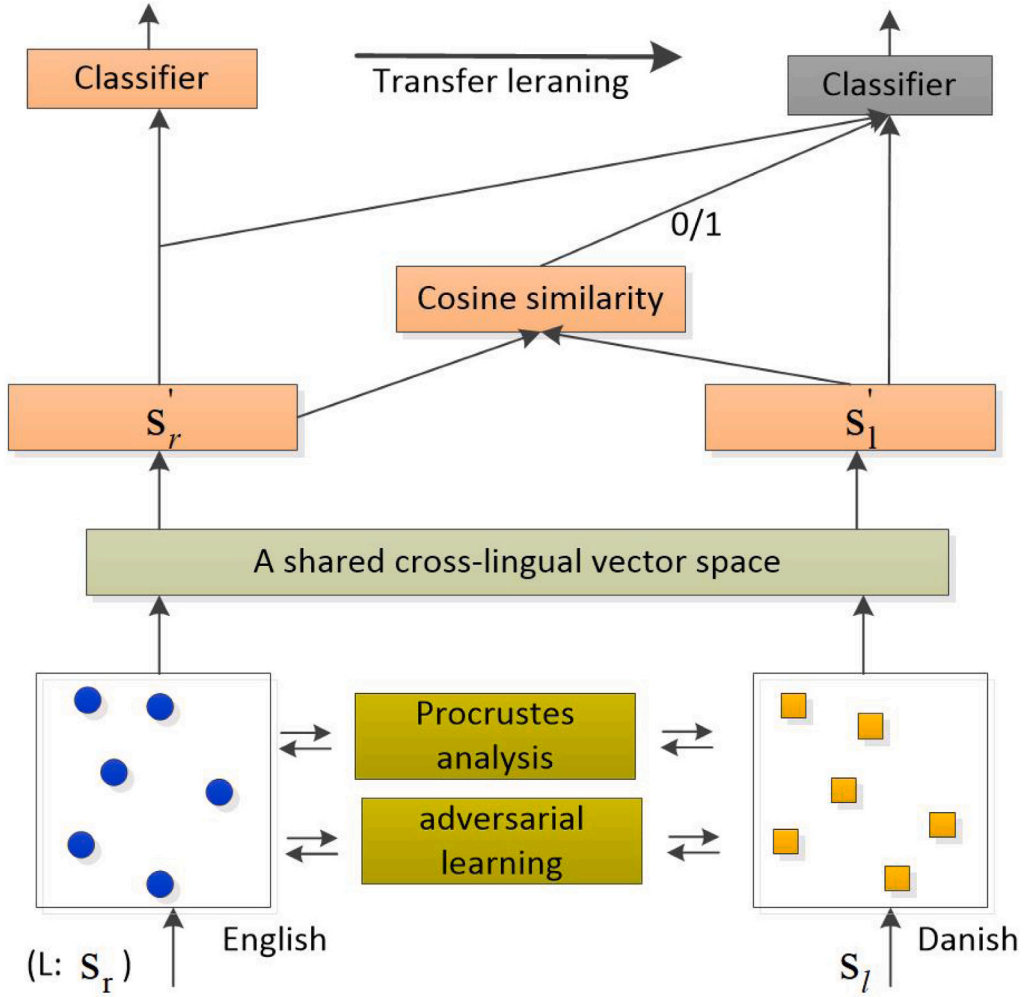


Fig. 1. Our proposed method that based on unsupervised transfer training for offensive speech detection.

3. Methodology

The overview of the model architecture is as shown in Fig. 1. Our proposed unsupervised offensive speech detecting is composed of three components: an unsupervised cross-lingual mapping, similarity-based sample generator, and agreement regularized transfer learning. In this paper, we use BERT(Bidirectional Encoder Representation from Transformers) to initialize word representations. Then, we propose to use adversarial training and Procrustes analysis to map two monolingual word embeddings into a shared space. For low-resource language offensive speech identification, we present a similarity-score method to generate some labeled data of low-resource language. Finally, we introduce an agreement regularized transfer learning that can hold the performance of the trained offensive speech identification model from rich-resource to low-resource languages.

3.1. Problem setup

Given monolingual text data (D_{rich} , D_{low}) from both the rich-resource and low-resource languages with a subset of labeled samples (D_{rich} , L_{rich}) of the rich-resource language where L_{rich} is a vector of class labels, the task of this paper is to build a universal classification model $f(S; \theta) \rightarrow L$ parameterized by θ that can be applied to unlabeled data for the low-resource language, where S is an input sentence from any language and L is its class label. In this paper, we assume rich-resource and low-resource languages should share the same class types.

Therefore, there are three crucial points to implement unsupervised cross-lingual offensive speech. The first is to construct shared word representations. Then the second is to train a classifier on rich-resource language. Finally, as different languages can be represented in a shared space, we can transfer the classifier that is trained on rich-resource language into a low-resource situation.

Based on the three points, we present our model (as shown in Fig. 1). It mainly contains: an unsupervised cross-lingual mapping, similarity-based sample generator, and agreement regularized transfer learning.

3.2. Unsupervised cross-lingual mapping

The main appeal of transfer learning is its potential to leverage models trained on data from outside the domain of interest. This can be particularly helpful for boosting the performance of learning low-resource languages. The recent success of cross-lingual pre-trained language model with transfer learning in offensive language identification for low resource languages [18] confirms that this is a feasible approach. The core of the cross-lingual pre-trained language model is to learn a shared language-invariant feature space between different languages. In this work, we first use an adversarial training method to implement the mapping process. The adversarial training mainly contains a generator and a discriminator. The generator G tries to make matrices of word representations to confuse the discriminator D . In this paper, we use S_r as source embeddings matrices and S_l as target embeddings matrices. At the same time, the discriminator D tries to distinguish the word representations which are from generator G or real-world. Besides, orthogonal parametrization of word representations can improve the mapping. To get orthogonal parametrization, we can find that transforming the source r word representations into the target, its transpose should be the target l to the source. In this paper, we use W as a word representation. The $D(\text{source} = 1|x)$ means that a representation W can be mapped into a source embedding. The generator can be formulated by minimization as follows:

$$L_G = -\log D(\text{source} = 0|W_r S_r) - \log D(\text{source} = 1|S_l) \quad (1)$$

The discriminator D aims to enhance its ability to distinguish S_l and $W_r S_r$. The discriminator can be implemented by maximizing as follows:

$$L_G = -\log D(\text{source} = 1|W_r S_r) - \log D(\text{source} = 0|S_l) \quad (2)$$

Although adversarial training can map different language word representations in a shared space, there is still a gap between different languages. To address this problem, we further propose Procrustes analysis to refine the mapping. In our English–Danish experiments, the average accuracy of refining is better than without refining process. Procrustes analysis is effective to align two shapes, [22] used it to align different faces. In this work, we employ the Procrustes analysis to refine our bilingual mapping. The process mainly contains two steps. To use the Procrustes analysis to align the two embedding spaces, we need to detect the suitable feature vectors of two embedding spaces as anchor points. As we are known, every language has some special function words, such as some stop words of English or “ya” “ha” “hei” in Chinese. Although some languages have given some special function words, there is always existing in low-resource language pairs. We propose to use the term frequency–inverse document frequency (TF–IDF) to filter anchor points. TF–IDF is a good idea in information retrieval to identify or remove such function words. We count the frequency of words in sentences. TF–IDF can be formularized as follow:

$$TD - IDF = \log \frac{N}{N(W_i)} \cdot (\log \frac{N+1}{N(W_i)+1} + 1) \quad (3)$$

N is the total number of words in the corpus, W_i is the number of i th words.

3.3. Similarity-based sample generator

After the above refining, the word embeddings should be closer to each other in this new embedding space, similar semantic sentences of different languages should have a higher score than dissimilar sentences. Based on this point, we propose a similarity-based sample generator method for low-resource languages. As there are many labeled data of rich-resource languages, we calculate the similarity score of sentences of rich-resource and low-resource languages to generate some candidate labeled data of low-resource language. In this paper, we compute the similarity score based on the margin between sentences of rich-resource and low-resource language. The margin-based similarity score calibrates the cosine similarity by dividing it by the average cosine distance of each word embedding's k nearest neighbor:

$$\text{score}(s_r, s_l) = \frac{\cos(s_r, s_l)}{\sum_{z \in NN_k(s_r)} (\frac{\cos(s_r, z)}{2k}) + \sum_{z \in NN_k(s_l)} (\frac{\cos(s_l, z)}{2k})} \quad (4)$$

s_r and s_l are word embeddings. $NN_k(s_r)$ denotes the k nearest neighbors of s in the other language excluding duplicates, and analogously for $NN_k(s_l)$. We mark sentences of low-resource language with margin scores above the pre-selected threshold as the same label with a labeled sentences of the rich-resource language. For example, the labeled data is very scarce or even none for Danish so that we do not have labeled data for training the detection model for Danish. Our method is to generate some candidate samples to train the model. In detail, we first calculate the similarity between a Danish sentence and a labeled English sentence. If the similarity is above the pre-selected threshold,⁷ we mark the Danish sentence as the same label as the labeled English sentence. Then, we use the generated samples in the next section.

⁷ In this paper, we set the threshold as 0.8 to hold the performance of our model.

Table 1
Datasets statistics.

Language	Train Set			Test Set		
	OFF	NOT	Total	OFF	NOT	Total
English	300K	300K	600K	1080	2807	3887
Danish	307	2061	2368	41	288	329
Turkish	4837	20184	25021	716	2812	3528
Arabic	1371	5468	6839	402	1598	2000
Greek	1989	5005	6994	242	1302	1544

3.4. Agreement regularized transfer learning

In this paper, we propose an unsupervised transfer learning to detect offensive speech of low-resource languages. In detail, we first train a good classifier on rich-resource language. For example, we train the classifier in English. For the classifier, many works design a lot of excellent classifiers [12,19]. In this paper, we mainly explain our proposed method and verify the effectiveness, so we use a simple feed-forward neural network based on LSTM with two hidden layers. Then, we train a classifier to match predicted labels with ground truth from the labeled data in rich-resource languages (such as English) as follows:

$$P(s) = \text{softmax}\left(\frac{e^{\text{enc}(\theta)}}{1 + e^{\text{enc}(\theta)}}\right)\epsilon(0, 1) \quad (5)$$

In above, we use θ denotes parameters of the classifier neural network $\text{enc}(\cdot)$. Then, we transfer the trained model into detecting offensive speech of low-resource languages. However, if we directly use the trained model to detect offensive speech of low-resource languages, the model will get a poor performance (we can find these points in Sec 5). In previous works, many methods use labeled data of low-resource languages to re-train the trained model of rich-resource languages. However, those methods are not suitable for our situation, as we generate some “fake” labeled data of low-resource languages. A basic point is that if the re-trained model is effective for low-resource languages after transfer learning, the model also is effective for rich-resource languages. Note that when we transfer the model into a new language, we need to share the hyper-parameters of the classifier neural network. Our agreement regularized transfer learning objective is to re-train the trained model with ground truth from the rich-resource language and “fake” labeled data of low-resource language and to encourage the hyper-parameters of the classifier neural network of two languages to be as similar as possible. Therefore, we use the following objective function:

$$L_c(\text{low}) = \mathbb{E}_{(x,y)}[\Delta(l, p(\theta)) + D_{\text{loss}}(\underbrace{\text{enc}[\theta_s] \parallel \text{enc}[\theta_s]}_{\min})] \quad (6)$$

where X is the representations in rich-resource languages, Y is the representations in low-resource languages. D_{loss} is minimizing the outputs of the classifier neural network of two languages (rich-resource and low-resource languages). By this, we can hold the performance of the model when performing transfer learning.

4. Experimental setting

In this section, we mainly present our experimental settings and describe the datasets used.

4.1. Dataset

To perform transfer learning, we used the aforementioned publicly available offensive speech detection datasets are presented in Table 1. For rich-resource language, we use the English large labeled dataset (called EN-OLID) to train our offensive speech detection model. The OLID (EN-OLID) is one of the most popular offensive language datasets for English. For other low-resource languages, we select four languages separately Danish, Greek, Arabic, and Turkish. To evaluate the quality of our proposed unsupervised offensive speech model, we need to compare it with a gold standard dataset for reference. In this paper, we utilize the provided labeled datasets by the organizers for Turkish [23], Danish [18], Greek [7], and Arabic [24] languages. The detailed statistics of the datasets are summarized in Table 1. From Table 1, we can find that the number of four low-resource languages (Danish, Turkish, Arabic, Greek) is very few, such as Danish only have two thousand training data. The data is seriously insufficient for training an offensive speech detection.

4.2. Baselines

- **BERT-SELF:** This is a basic baseline. We directly use the few training data of low-resource language to train the classifier model.
- **BERT&EN-SELF [16]:** They took advantage of the English data available by applying cross-lingual BERT and transfer learning to make predictions in languages with low-resource.
- **Augmentation [10]:** They proposed a cross-lingual augmentation approach that can adapt and fine-tune the BERT and Multilingual BERT models for low-resource language.
- **Network Embedding [8]:** They proposed a weak supervision model that can address a specific gap by employing only very few lexicon labels to determine hate speech.

Table 2
Results of our proposed systems on a shared training set.

Language	model	Accuracy	MacroPrecision	MacroRecall	F1
Danish	BERT-SELF	0.581	0.433	0.448	0.44
	BERT&EN-SELF	0.878	0.729	0.749	0.739
	Augmentation	0.912	0.803	0.758	0.780
	Network Embedding	0.912	0.803	0.758	0.780
	Ours	0.932	0.859	0.806	0.829
Arabic	BERT-SELF	0.596	0.523	0.463	0.491
	BERT&EN-SELF	0.852	0.756	0.793	0.774
	Augmentation	0.895	0.841	0.828	0.834
	Network Embedding	0.901	0.845	0.832	0.838
	Ours	0.926	0.88	0.863	0.871
Turkish	BERT-SELF	0.582	0.593	0.603	0.598
	BERT&EN-SELF	0.802	0.789	0.732	0.759
	Augmentation	0.820	0.836	0.728	0.778
	Network Embedding	0.843	0.801	0.746	0.773
	Ours	0.881	0.842	0.798	0.819
Greek	BERT-SELF	0.589	0.569	0.493	0.528
	BERT&EN-SELF	0.789	0.756	0.7683	0.759
	Augmentation	0.830	0.769	0.788	0.775
	Network Embedding	0.852	0.838	0.801	0.819
	Ours	0.878	0.843	0.811	0.827

4.3. Settings

To better train the model, we will use a 5-layer BILSTM with 512 dimensions per layer and a sentence length control of 80. Since the model uses a bidirectional LSTM, the dimension of the final generated sentence vector representation is 1024. The initial learning rate is set to 0.001 and the model will be trained for 100 epochs.

For low-resource model training, it is necessary to rely on models trained in rich-resource languages. To better perform transfer learning on low-resource languages, first, use the sample data set composed of rich languages to train the model, and then use a new training set composed of the rich language and the “fake” low-resource samples labeled according to the marginal score with the rich language for the transfer learning of the model. This not only solves the problem of insufficient training sets in low-resource language transfer learning; but also enables the model parameters to be fine-tuned as much as possible to ensure the best effects of transfer learning.

5. Results and evaluation

In this section, we give the results of baselines and our method. We first present the main results, then test each setting to how affects offensive speech detection of low-resource languages.

5.1. Overall performance

In our experiments, we use Accuracy,⁸ MacroPrecision(P),⁹ MacroRecall(R)¹⁰ and MacroF₁-score(F₁)¹¹ to evaluate our method.

From Table 2, the F₁ score of our model is 0.829 higher than BERT-SELF. This means when the number of training data is very, the classifier model cannot obtain a good performance. The other evaluation indexes have the same results. The method of BERT&EN-SELF [16] uses the model trained on rich-resource language to transfer into low-resource language. Then, they transfer the trained model and adjust the model using labeled data of low-resource language. We can find the results of BERT&EN-SELF are better than BERT-SELF. Their method only try to adjust the parameters of a classifier that is trained on rich-resource language, they do not refine the cross-lingual word representations when they implement transfer learning.

The methods of augmentation [10] and Network Embedding [8] (also can be called weakly supervised methods) all attempt to adjust the parameters of the classifier when they implement transfer learning in low-resource languages. We can find that they all obtain better results than the first two baselines. In addition, the results them are similar in some languages. However, they all ignore language-invariant representations of language when they implement transfer learning. A strong point is that the refined model after transfer learning still have a good performance in rich-resource languages. They only refine one language when they

⁸ Accuracy represents the ratio of the number of correctly classified samples in the prediction result to the number of all samples.

⁹ MacroPrecision represents the ratio of the true positive samples in the prediction result to all the positive samples in the prediction result.

¹⁰ MacroRecall represents the ratio of the true positive samples in the prediction result to the original positive samples in the sample.

¹¹ The F1 score is used as the main basis for comparing experimental methods. Since the F1 score is the harmonic mean of precision and recall, the harmonic mean is greatly affected by extreme values, and is more suitable for evaluating the classification problem of unbalanced data. The F1 score represents the ratio of 2 times the product of MacroPrecision and MacroRecall to the sum of MacroPrecision and MacroRecall.

Table 3

The results are that the cross-lingual mapping how to affect the transfer learning.

Language	model	Accuracy	MacroPrecision	MacroRecall	F1
Danish	NON	0.101	0.091	0.082	0.086
	ADV	0.867	0.731	0.747	0.742
	ADV+P	0.932	0.859	0.806	0.829
Arabic	NON	0.092	0.085	0.081	0.083
	ADV	0.886	0.782	0.813	0.794
	ADV+P	0.926	0.88	0.863	0.871
Turkish	NON	0.082	0.093	0.083	0.098
	ADV	0.792	0.769	0.721	0.745
	ADV+P	0.881	0.842	0.798	0.819
Greek	NON	0.088	0.086	0.079	0.077
	ADV	0.793	0.761	0.783	0.778
	ADV+P	0.878	0.843	0.811	0.827

adjust the parameters of the classifier. This method may cause the model of low-resource language to be different from rich-resource language. For our method, we present two strategies to improve the performance of offensive speech in low-resource languages. At the same time, our method is unsupervised so that it is very suitable for low-resource languages. For Table 2, the performance of our unsupervised method either is on par or outperforms the supervised methods.

5.2. The effect of cross-lingual mapping for transfer learning

We all know that the basics of transfer learning the word shared representations in natural language processing applications. In Table 3, we mainly test the cross-lingual mapping to how to affect transfer learning.

The “NON” means when we directly use the model that trained rich-resource language to detect offensive speech of low-resource language. From the Table 3, we can find that using the model trained on rich-resource language to detect low-resource language offensive speech is not enough without mapping different word representations into a shared space. Then, we use “ADV” to stand for that we directly transfer the trained model into low-resource language with adversarial learning. We can find the result is very better than “NON”. This means mapping different word representations into a shared space is very important when we implement transfer learning. However, only using adversarial learning may mistake some word alignment. Therefore, we add Procrustes analysis after adversarial learning (ADV+P) to refine the mapping. We can find the refining process is very useful for transfer learning. From Table 3, we can watch that the results of “ADV+P” are the best for four languages. We use adversarial learning and Procrustes analysis to implement simultaneously the mapping on two languages. The Procrustes analysis can obtain better cross-lingual word representations so that the trained model can have better results in low-resource languages.

5.3. The size of training data for transfer learning

As we all know, we also need to retrain the trained model on a new dataset when we implement transfer learning. Therefore, we implement an experiment about how the size of training data of low-resource language affects the performance of transfer learning. It can be found from Fig. 2 that the number of datasets involved in model training will affect the model training effect. Especially when the number of participants in training is greater than 3000, the increase in model effect tends to be stable, but still maintains an upward trend. For example, When the amount of data reaches 3000, the accuracy of Greek remains relatively stable, and increasing the amount of training data will not greatly improve the effectiveness of the model. The number of other languages participating in training and the model training effect is shown in Fig. 2. Combining Table 1, we can find that Danish is a very low-resource language. For example, the number of Danish data is less than 2,400 that contains positive and negative samples. This means that we cannot obtain a good result for Danish as there is not enough training data for transfer learning. To address this problem, we propose to use the cross-lingual word embeddings to generate some samples to improve the performance of low-resource languages. We will show the results of our generator in the next section.

5.4. The effect of sample generator for transfer learning

This section mainly discusses the impact of sample generators on transfer learning. In this paper, we use cross-lingual word embeddings to generate some candidate training data for low-resource languages. Then, we use those generating data to refine the transferred model for low-resource languages. Table 4 gives the results.

In Table 4, the “NON” is that we directly use the model that trained on rich-resource languages to detect offensive speeches on low-resource languages. We can find the result is worst compared with the others. This means adjusting parameters for low-resource language is very important when we implement transfer learning. Then, we use “TEST” to represent that we use the training data provided by Table 1 to update the parameters of the classifier when we implement transfer learning. We can find that refining the transferred model on new training data is very useful for transfer learning. Finally, we use our generated training data to implement the process of updating the parameters of the classifier. We can find that the sample generator is effective for detecting offensive speeches. Noted that we do not use the labeled training data of low-resource languages, so it can be demonstrated the effectiveness of our generator.

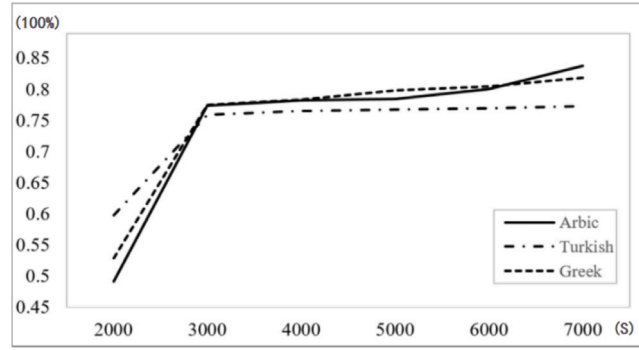


Fig. 2. The size of different training data of low-resource language is how to affect the performance of transfer learning. The x-axis is the percentage accuracy and the y-axis is the number of training sentences.

Table 4

Results of the sample generator how to affect the transfer learning.

Language	model	Accuracy	MacroPrecision	MacroRecall	F1
Danish	NON	0.461	0.383	0.418	0.438
	TEST	0.883	0.759	0.769	0.742
	GENERATOR	0.932	0.859	0.806	0.829
Arabic	NON	0.526	0.503	0.473	0.482
	TEST	0.862	0.763	0.801	0.782
	GENERATOR	0.926	0.88	0.863	0.871
Turkish	NON	0.556	0.507	0.473	0.468
	TEST	0.815	0.796	0.743	0.761
	GENERATOR	0.881	0.842	0.798	0.819
Greek	NON	0.563	0.51	0.481	0.475
	TEST	0.809	0.765	0.793	0.788
	GENERATOR	0.878	0.843	0.811	0.827

Table 5

Ablation study.

Language	model	Accuracy	MacroPrecision	MacroRecall	F1
Danish	NON	0.461	0.383	0.418	0.438
	ONE	0.919	0.842	0.738	0.777
	OUR	0.932	0.859	0.806	0.829
Arabic	NON	0.526	0.503	0.473	0.482
	ONE	0.918	0.866	0.847	0.856
	OUR	0.926	0.88	0.863	0.871
Turkish	NON	0.556	0.507	0.473	0.468
	ONE	0.874	0.825	0.761	0.791
	OUR	0.881	0.842	0.798	0.819
Greek	NON	0.563	0.51	0.481	0.475
	ONE	0.874	0.867	0.826	0.842
	OUR	0.878	0.873	0.861	0.867

5.5. Ablation study

In this paper, we propose simultaneously the regularized training that updates the parameters of the transferred model on two languages. To understand the effect of the regularized training for transfer learning in our model on the overall performance, we conduct an ablation study. The results are in Table 5.

In Table 5, we use “NON” to stand for that we directly transfer into low-resource language without adjusting parameters. We can find the result is worst. This means adjusting parameters for low-resource language is very important when we implement transfer learning. Then, we use “ONE” to represent that we only update the parameters of the transferred model on low-resource languages using training data. We can find that refining the transferred model for new tasks is very useful when we use transfer learning. Then, we implement simultaneously adjusting the parameters of the transferred model on two languages to demonstrate our agreement regularized training. We can observe that our agreement regularized training obtains the best results. Our agreement regularized training can make the transferred model keep the commonality of language. Therefore, our model can obtain the best results.

6. Conclusion

In the era of network prevalence, a large number of offensive languages appear in the network, and the current method for identifying cross-lingual offensive languages for low-resource needs to be improved. Therefore, this paper proposes an unsupervised attack detection method without any low resource language tag data. Facing the problem that the training set of low resource language is insufficient in the implementation of transfer learning, which will reduce the training effect of the model, we directly transfer the data set composed of rich language and “false” marked low resource language to the model trained with rich resource language. This method not only ensures the number of low resource training sets; At the same time, fine-tunes the model parameters as much as possible to ensure that the training effect of the model is more satisfactory.

We also prove the effectiveness of this method through a series of related experiments. For example, compare the experimental results of different models, the experimental results of different recognition methods, and the experimental results of different numbers of training sets. Among the Danish recognition results, the F1 value of our training equipment is 4.9 percentage points higher than that of other best equipment; Moreover, based on the protocol rules we proposed, the experimental results show that in the Danish recognition results, the F1 value is 5.13 percentage points higher than that of other protocols. Experimental results on four low resource languages show that our model is equivalent to or better than the supervised method without relying on any annotation data.

Declaration of competing interest

No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to <https://doi.org/10.1016/j.compeleceng.2022.108005>.

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